

Concerted Evolution of Vertebrate *CCR2* and *CCR5* Genes and the Origin of a Recombinant Equine *CCR5/2* Gene

ANDREY A. PERELYGIN, ANDREY A. ZHARKIKH, NATALIA M. ASTAKHOVA, TERI L. LEAR, AND MARGO A. BRINTON

From the Department of Biology, Georgia State University, PO Box 4010, Atlanta, GA 30302 (Perelygin, Astakhova, and Brinton); the Bioinformatics Department, Myriad Genetics, Inc., Salt Lake City, 320 Wakara Way, UT 84108 (Zharkikh); and the Department of Veterinary Science, University of Kentucky, 108 Maxwell H. Gluck Equine Research Center, Lexington, KY 40546 (Lear).

Address correspondence to Andrey A. Perelygin at the address above, or e-mail: aperelygin@gsu.edu.

Abstract

Chemokine receptors (CCRs) play an essential role in the initiation of an innate immune host response. Several of these receptors have been shown to modulate the outcome of viral infections. The recent availability of complete genome sequences from a number of species provides a unique opportunity to analyze the evolution of the *CCR* genes. A phylogenetic analysis revealed that the *CCR2* gene evolved in concert with the paralogous *CCR5* gene, but not with another paralogous gene, *CCR3*, in the opossum, platypus, rabbit, guinea pig, cat, and rodent lineages. In addition, evidence of concerted evolution of the *CCR2* and *CCR5* genes was observed in chicken and lizard genomes. A unique *CCR5/2* gene that originated by unequal crossing over between the *CCR2* and *CCR5* genes was detected in the domestic horse. The *CCR2*, *CCR5*, and *CCR5/2* genes were mapped to ECA16q21 using fluorescent in situ hybridization (FISH). Single-nucleotide polymorphisms identified in the equine *CCR5* gene and characterized within 5 horse breeds provide haplotype markers for future case/control studies investigating the genetic bases of horse susceptibility to infectious diseases.

The chemokine family includes a group of secreted proteins that activate the immune system during inflammation in response to microbial infection and during autoimmune or allergic reactions (Howard et al. 1996). Chemokines act on immune cells by binding to chemokine receptors (CCRs). These receptors are members of the G protein-coupled receptor family and are composed of 7 hydrophobic transmembrane (TM) domains as well as an extracellular N-terminal region and a cytoplasmic C-terminal tail that contains important structural and functional motifs. In addition to binding chemokine ligands, particular CCRs have previously been reported to facilitate immunodeficiency virus entry (Deng et al. 1996; Dragic et al. 1996; He et al. 1997; Lim et al. 2006).

Subcutaneous inoculation of West Nile virus (WNV) consistently led to a fatal outcome in *CCR5* knockout mice (Glass et al. 2005). The data obtained with this animal model suggested the possibility that the *CCR5Δ32* missense mutation, that provides protection against human immunodeficiency virus 1, might increase the risk of fatal WNV infection in humans. Cohort studies using clinical samples

from WNV-infected and control individuals from Arizona and Colorado showed that 1% of the controls but 4–8% of individuals with laboratory-confirmed, WNV-induced disease were homozygous for the *CCR5Δ32* allele. However, this difference was not statistically significant. An increased frequency ($P = 0.04$) of *CCR5Δ32* homozygotes among fatal cases in the Arizona cohort was reported to be statistically significant but only a few fatal cases were studied (Glass et al. 2006).

WNV is a mosquito-borne, single-stranded, positive-sense RNA virus that has recently become endemic in the Americas. WNV is maintained in a mosquito–bird–mosquito transmission cycle but humans and other mammals are occasionally infected. WNV infection does not cause severe disease and death in domestic animals, such as cattle (*Bos taurus*), pigs (*Sus scrofa*), rabbits (*Oryctolagus cuniculus*), cats (*Felis catus*), or dogs (*Canis lupus familiaris*). In contrast, horses (*Equus caballus*) and other equids that develop clinical WNV-induced disease show mortality rates up to 30% (Salazar et al. 2004; Schuler et al. 2004; Ward et al. 2006).

The mRNA sequences and exon/intron structures of the equine *CCR2* and *CCR5* genes and a unique *CCR5/2* gene were characterized in this study. These genes were FISH mapped to ECA16q21. Single-nucleotide polymorphisms (SNPs) were identified in the coding regions of the *CCR5* and *CCR5/2* genes. These SNPs will be utilized in future cohort studies to analyze associations between innate immunity genes and susceptibility to infectious disease in horses. Sequences of orthologous elephant and chicken *CCR5* genes were also determined. These sequences as well as *CCR2*, *CCR3*, and *CCR5* gene sequences from other species detected in the GenBank database were used for phylogenetic analysis. The results of this analysis provided evidence for concerted evolution of the vertebrate *CCR2* and *CCR5* genes.

Materials and Methods

RNA and DNA Sources

Leghorn chicken intestine full-length cDNA was purchased from Seegene (Rockville, MD). Both equine total RNA and genomic DNA were isolated from Quarter horse peripheral white blood cells (PWBC) kindly provided by Dr T. Thompson (Equine Medicine and Surgery, Douglasville, GA). This equine PWBC RNA as well as spleen total RNA that was isolated from a horse of an unknown breed (Zyagen Laboratories, San Diego, CA) were converted into single-stranded cDNA (sscDNA) as described previously (Perelygin et al. 2005). High molecular weight genomic DNA isolated from an African savannah elephant was kindly provided by Drs A. Roca and S. O'Brien (National Cancer Institute, Frederick, MD).

Extension of Partial Gene Sequences

The DNA Walking (DW) SpeedUp kit (Seegene) was used to extend partial gene sequences. This kit utilized both annealing control primer (ACP) and dual priming oligonucleotide technologies, which maximize the ability of commercial DW-ACP primers and target-specific primers to capture unknown target sites and optimize polymerase chain reaction (PCR) conditions. Three gene-specific primers are required for each walking step, which significantly increases the specificity of each of the partial sequence extensions. In contrast to the rapid amplification of cDNA ends method, specific adapter ligation is not required and both genomic DNA and cDNA templates can be used with the DNA walking SpeedUp kit.

Detection of Polymorphisms in the Equine *CCR5* and *CCR5/2* Genes

To identify SNPs in the equine *CCR5* and *CCR5/2* genes, genomic DNA samples from 1 Arabian, 1 Hanoverian, 1 Paint, 4 Quarter, 3 Thoroughbred, and 6 mix-breed horses were isolated as described previously (Perelygin et al. 2006a) and amplified by PCR using the primers listed in Table 1. Because of high similarity between *CCR2* and *CCR5*

Table 1. PCR primers used to amplify vertebrate *CCR2*, *CCR5*, and *CCR5/2* genes

Species	Primer name	Primer sequence
Horse	EcCCR5degF	5'-cttctcatcatcctcytgac-3'
	EcCCR5degR	5'-ttgatgcagcagtgycat-3'
	EcCCR5rnaF	5'-tgattcacacagctcagcagagca-3'
	EcCCR5rnaR	5'-aaccacaacttgagggggaaatc-3'
	EcCCR5dnaF	5'-ctttactcacagggttgagcaag-3'
	EcCCR5dnaR	5'-aaccacaacttgagggggaaatc-3'
	EcCCR2rnaF	5'-tcgtaggcacaggagcagatg-3'
	EcCCR2rnaR	5'-tcagcaaaccccgagggtgtactat-3'
	EcCCR5/2rnaF	5'-tgattcacacagctcagcagagca-3'
	EcCCR5/2rnaR	5'-tcagcaaaccccgagggtgtactat-3'
	EcCCR5/2dnaF	5'-ctttactcacagggttgagcaag-3'
	EcCCR5/2dnaR	5'-tcagcaaaccccgagggtgtactat-3'
Elephant	LaCCR5dnaF	5'-ctagtgtcgagagcttagccgaa-3'
	LaCCR5dnaR	5'-cctctgtggcagtaacatccctt-3'
Chicken	GgCCR5rnaF	5'-ccaagtcaccactgtcaccttatg-3'
	GgCCR5rnaR	5'-gtgtgcatgacttttcgagatg-3'

sequences, special precautions were taken to ensure that only the correct gene was amplified. First, long primers (usually 24 mers) with high annealing temperatures (68–72 °C) were used to amplify specific PCR products. Second, each primer pair was designed from gene-specific sequences so that each amplified only 1 *CCR* gene. Third, all PCR amplifications were performed using the Advantage 2 polymerase mix (Clontech, Mountain View, CA), which contains Taq-polymerase antibody that provides high PCR specificity compatible with “hot-start” PCR. PCR products were individually purified using a QIAquick PCR Purification Kit (Qiagen, Valencia, CA) and directly sequenced using the BigDye Terminator v1.1 Cycle Sequencing Kit (Applied Biosystems, Foster City, CA) in the DNA/Protein Core facility at Georgia State University.

SNP Genotyping within the Equine *CCR5* Gene

The tetraprimer PCR amplification of refractory mutation system (tetraprimer ARMS-PCR) technique (Ye et al. 2001) or the restriction fragment length polymorphism (RFLP) technique were used to genotype identified SNPs within the equine *CCR5* gene in 5 horse breeds: American Saddlebreds, Arabians, Quarter horses, Standardbreds, and Thoroughbreds. The PCR primer pairs used for genotyping are listed in Table 2.

Phylogenetic Analysis

CCR2 and *CCR5* phylogenetic trees were built using the njtree program with 2-parameter (Kimura 1980) or codon-based (Li 1993) distances as described previously (Perelygin et al. 2006b). This program is available on request from A.A.Z. (zharkikh@myriad.com).

FISH Mapping of Equine *CCR5*

The horse CHORI-241 BAC library was searched with a probe derived from the partial equine *CCR5* cDNA that

Table 2. PCR primers and restriction enzymes used to genotype SNPs within the ORF of the equine *CCR5* gene in DNA samples from 5 breeds of horses

SNP	Inner primer pairs or restrictase	Outer or regular primer pairs
A63G	5'-cgattacggtatgdtcagagccctgcaaa-3' 5'-ctgcgattgtctcacgtcggttctc-3'	5'-tagtgagcagaaaagctggcagcaaac-3' 5'-aagcaagtcagagatggccaagttgagc-3'
C108G	BsrBI	5'-gttggccaaaaaaggagctca-3' 5'-tccaagctccactgggtgagc-3'
C267T	HpyCH4V	5'-cgacaagtcggttctatgacatcg-3' 5'-aaggtgaccgtcctgcttttaaa-3'
C399T	5'-acaggtacctggctatcgtccattcc-3' 5'-cgctcctggcttttaagcaaaccca-3'	5'-gcctcgtcctgataaagtgcaaa-3' 5'-tccagaaatggtactgactgggtggg-3'

was obtained as part of this study using degenerative primers EcCCR5degF and EcCCR5degR (Table 1). DNA from equine BAC clones was labeled with Spectrum Orange-dUTP (Vysis, Downer's Grove, IL) following the manufacturer's directions. These DNAs were FISH mapped on equine metaphase spreads as described previously (Perelygin et al. 2006a).

Results and Discussion

Assembling the Horse, *CCR2*, *CCR5*, and *CCR5/2* Genes

Cat (AB022910), cattle (AY834252), human (NM_000579), pig (AB119272), and rabbit (DQ444458) *CCR5* mRNA sequences were downloaded from GenBank and aligned to design degenerative primers (EcCCR5degF and EcCCR5degR; Table 1) located in evolutionarily conserved regions. Using these primers, 3 fragments were amplified from equine PWBC sscDNA by PCR. The sequence of one of these fragments was similar to those of other mammalian *CCR5* genes. This cDNA sequence was extended to full length by DNA walking and submitted to GenBank under accession number DQ629590. Alignment of this sequence with the horse whole-genome sequence assembly from the UCSC genome browser (www.genome.ucsc.edu) revealed 2 exons: the noncoding exon 1 of 285 bp and exon 2 that includes a 5' NCR of 11 bp, an open reading frame (ORF) of 1059 bp, and a 3' NCR of 679 bp (Figure 1). The exon 1 (5' NCR) and exon 2 (3' NCR) sequences may not be complete due to limitations of the DNA walking method. The ORF and 3' NCR of the DQ629590 sequence showed 99–100% similarity to several sequences (EF507698 through EF507703) derived from Portuguese horse genomic DNAs and recently submitted to GenBank by van der Loo and coworkers, Research Center in Biodiversity and Genetic Resources, Vairao, Portugal. No alternatively spliced variants of the horse *CCR5* mRNA were detected after amplification of either equine PWBC or spleen sscDNA with horse *CCR5* gene-specific primers (EcCCR5rnaF and EcCCR5rnaR; Table 1). Although only one version of the horse *CCR5* mRNA sequence (DQ629590) has been detected to date (Figure 1), 2 alternatively spliced variants of the human *CCR5* mRNA were previously reported (Mummidi et al. 1997).

The sequence of another of the PCR fragments was similar to both horse expressed sequences tag (EST) CD466909 and the whole-genome contig AAWR01020520. This sequence represented a middle portion of the equine *CCR2* gene. A similar genomic sequence consisting of the

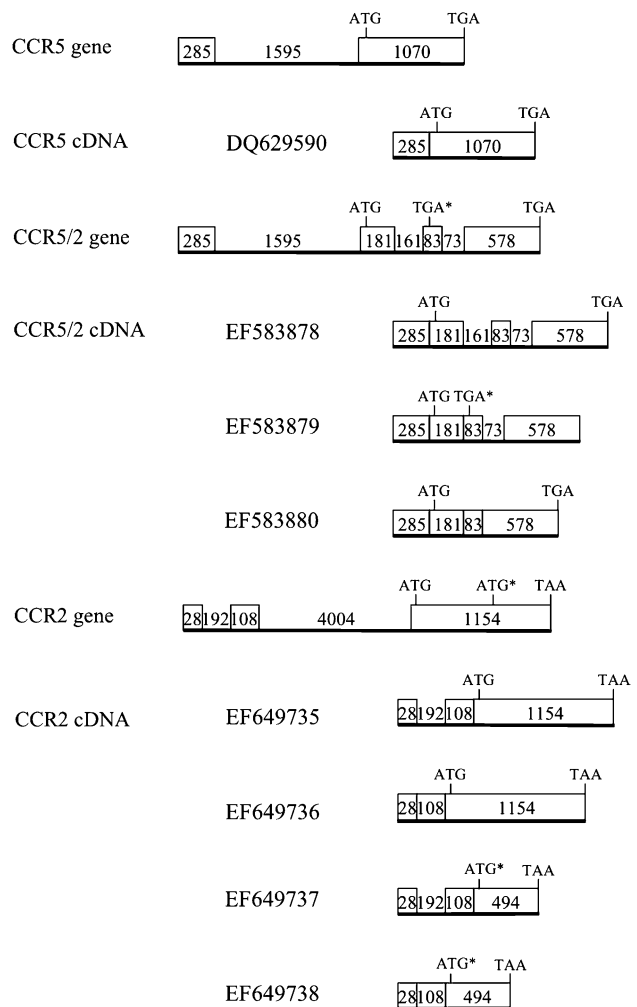


Figure 1. Splice variants of equine *CCR5*, *CCR5/2*, and *CCR2* transcripts. Asterisks indicate cryptic translation initiation or termination codons. The 3' NCRs were not included for any of the genes.

3' end of intron 1 and the ORF of the horse *CCR2* gene was recently submitted to GenBank by van der Loo and coworkers (DQ473316). Alignment of EST CD466909 with the DQ473316 and AAWR01020520 sequences (data not shown) revealed a 660-bp gap in the middle of the CD466909 sequence that suggested the existence of an alternatively spliced version of the equine *CCR2* mRNA. To further analyze this possibility, equine spleen ssDNA was amplified using *CCR2*-specific primers (EcCCR2rnaF and EcCCR2rnaR; Table 1). Four alternatively spliced transcripts of various lengths were detected. Their sequences were determined and submitted to GenBank under accession numbers EF649735 through EF649738. Alignments of each of these cDNA sequences with the sequence of the genomic contig AAWR01020520 revealed 3 alternatively spliced exons in the equine *CCR2* gene (Figure 1). A GenBank search using the BLASTN program indicated that the 5' portion (328 bp) of the largest equine *CCR2* transcript (EF649735) is similar to the 5' noncoding exon of the recently reported porcine ortholog NM_001001619 (Shinkai et al. 2005). The EF649736 sequence, which is 99% identical to the Portuguese DQ473316 sequence, was 192 bp shorter than the EF649735 sequence, suggesting the presence of 2 noncoding exons at the 5' end of the equine *CCR2* gene (Figure 1). Two noncoding 5'-terminal exons were also detected in the murine ortholog by searching the UCSC genome browser. The EF649737 version of equine *CCR2* mRNA differed from EF649735 by the lack of 660 bp at the 5' end in the single 3'-terminal exon due to alternative splicing. The splicing of both introns as well as of the 660-bp region at the 3'-terminal exon generated the EF649738 version of equine *CCR2* mRNA (Figure 1). Both EF649735 and EF649736 mRNAs encode the full-length equine *CCR2* protein, whereas both EF649737 and EF649738 mRNAs encode a truncated protein. Alternative splicing was previously reported to occur at the 3'-end of human *CCR2* pre-mRNA (Charo et al. 1994).

A GenBank search using the sequence of the third amplified equine PCR fragment revealed that its 5' portion was similar to other mammalian *CCR5* gene sequences, whereas its 3' portion was similar to orthologous *CCR2* gene sequences as well as to the equine genomic contig AAWR01020518, suggesting that a unique *CCR5/2* gene had originated in the horse genome by unequal crossing over between the *CCR5* and *CCR2* genes (Figure 2A). When the equine PWBC ssDNA was amplified using *CCR5/2* specific primers (EcCCR5/2rnaF and EcCCR5/2rnaR; Table 1), 3 cDNAs of various lengths were detected and sequenced. The sequences of these cDNAs were submitted to GenBank under accession numbers EF583878 through EF583880. Alignment of these 3 cDNA sequences with the AAWR01020518 genomic contig revealed 3 introns in the equine *CCR5/2* gene (Figure 1). One, 2, or all 3 of these introns were spliced out in the EF583878, EF583879, and EF583880 *CCR5/2* mRNAs, respectively. A different protein would be translated from each of the 3 alternatively spliced *CCR5/2* mRNAs. Sequences of orthologous *CCR5/2* genes from donkey (*Equus asinus*) and Grevy's zebra (*Equus*

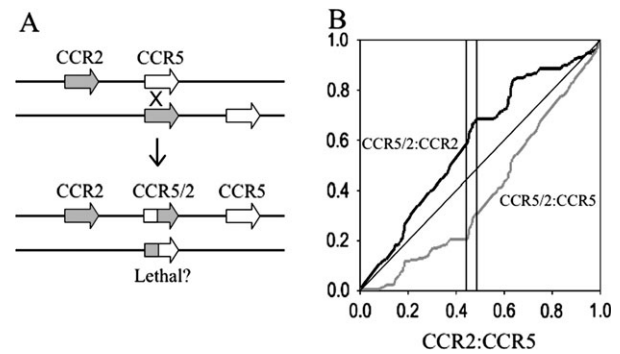


Figure 2. The origin of the horse *CCR5/2* gene. (A) Model of unequal recombination between the horse *CCR5*–*CCR2* chromosomal regions. Arrows indicate direction of RNA transcription. (B) Phase diagram of the cumulative distributions of differences between the horse *CCR* genes: the horizontal axis corresponds to the cumulative difference between the *CCR2* and *CCR5* genes; the vertical axis corresponds to the cumulative difference between the *CCR5/2* and *CCR2* (black line) or between *CCR5/2* and *CCR5* (gray line) genes. Identical distributions would follow the diagonal line. The vertical lines indicate the interval in which the recombination event occurred.

grevyi) were recently submitted to GenBank by van der Loo and coworkers (EF515830 and EF515831, respectively). These 2 sequences showed 98–99% similarity to the orthologous EF583880 *CCR5/2* horse sequence but only about 90% similarity to either the DQ629590 *CCR5* or the EF649736 *CCR2* paralogous horse sequences.

Mapping of Horse *CCR2*, *CCR5*, and *CCR5/2* Genes

Two BAC clones, 4714 and 85J10, were found to contain the equine *CCR2*, *CCR5*, and *CCR5/2* genes. The coding exons of the *CCR5* and *CCR5/2* genes as well as the 3' portion of the *CCR2*-coding exon were PCR amplified from each of these 2 BAC clones and sequenced. The sequences obtained showed complete identity with the corresponding cDNA sequences described above. FISH mapping with these 2 BAC clones assigned the *CCR2*, *CCR5*, and *CCR5/2* genes to the horse chromosomal location ECA16q21 (data not shown). This map position was in agreement with the positions of the *CCR2* and *CCR5* genes on the horse genome sequence assembly (UCSC genome browser).

SNPs in Coding Regions of the Horse *CCR5* and *CCR5/2* Genes

Amplification and direct sequencing of DNA from 16 unrelated horses of various breeds using *CCR5* specific primers (EcCCR5dnaF and EcCCR5dnaR; Table 1) identified 4 variable positions in the equine *CCR5* gene. All of these SNPs (A63G, C108G, C267T, and C399T; nucleotide numbering started from the translation initiation site in the cDNA sequence) were synonymous. The EcCCR5/2dnaF and EcCCR5/2dnaR primers (Table 1) were used to identify

CCR5 ATGGATTATCAGACGACAAGTCCGTTCTATGACATCGATTACGGTATGTCAGAGCCCTG
CCR5/2 ATGGGTTATCAGACGACAAGTCCGTTCTATGACATCGATTACAGT**AM**GTCAAGGCCCTG

CCR5 CCA**RA**AAACCGACGTGAGACAAATCGCAGCCAGGCTCCTGCCCCGCT**ST**ACTCGCTGG
CCR5/2 CCA**AA**AAACCGA**Y**GTGAGACAAATCGCAGCCAGGCTCCTGCC**M**CCGCTCTACTCGCTGG

CCR5 TGTTCATCTCTGGTTTGTGGGCAACTTGTTAGTCGCCCTCGTCTGATAAAGTGCAAA
CCR5/2 TGTTCATCTCTGGTTCTCTGGGCAACATGCTGGTCATCCTTGCTCTGATAAAGTCA**RTA**

CCR5 AAGCTGAAGAGCATGACTGACATCTACCTGCTCAACTTGGCCATCTCTGACTTGCTTTT
CCR5/2 AAGCTAAAGAGGGTGGCTGACATCTACCTGCTCAACTTGGCCATCTCTGACTTGCTTTT

CCR5 CCTTCTCACCTCCCATTTCTGGGCTCACTA**Y**GCAGCTCACCAGTGGGAGCTTGGAATA
CCR5/2 CGTTCTCACCTCCCATTTGTGGGCTCACTACGCAGCACACAGTTGGGTTTTTGGAAT**R**

CCR5 CAATGTGTCGACTTCTGACTGGGCTCTATTTATAGGCTTCTTCTCTGGAATCTTCTTC
CCR5/2 GAATGTGTCAACTTTCGATTGGACTCTATTTATAGGCTTCTTCTCTGGAATCTTCTTC

CCR5 ATCATCCTCTGACAATCGACAGGTACCTGGCTATCGTCCATGC**Y**GTGTTTGCTTTAA
CCR5/2 ATCATCCTCTGACAATCGACAGGTACCTGGCTATCGTCCATC**ST**GTGATTCCTTTAA

CCR5 AGCCAGGACGGTCACCTTTGGGCTGATGACAAGTGGGGTCACTTGGGCGGTGGCTGTGT
CCR5/2 GGTGAGCACCGTCGCCCTTTGGGGTGGTGCAAGTGGGGTCACTTGGCTGGTG**R**CTGTGT

CCR5 TTGTCTCTCTCCAGGAATCATCTTTACCAAATCCCAAAAGAGGGTACTCGTTTTACA
CCR5/2 TTGCCTCTCTCCAGGAATCATCTTTACCAAATCCCAAAAGAACTTCTTGGAACTCT

CCR5 TGCAGCCCTCATTACCCACCCAGTCAGTACCATTTC-----TGGAGGAATTCCA
CCR5/2 GAGAAAGAGTCTGTTTACTCCTGCGGCCCTTATTTCCACCACAATGGAGGAATTCCA

CCR5 GGCATTAAAGATGACCATCTTGGGCTGGTCTGCCACTGCTGGTCATGATCATCTGCT
CCR5/2 TATAATAATGATCACCATTCTCAGCCTGGTCTGCCACTGCTGGTCATGATCATCTGCT

CCR5 ACTCCGGAATCCTGAAAACCTGCTTCGGTGTGCAACGAGAAGAAGAGGCACAAGGCC
CCR5/2 ACTCAGCAAT**I**CTGAAAACCTGCTTCA**RT**GTCTCCCCAGGAAGAAG---CATAAGGCC

CCR5 GTGAGGCTCATTTTCGTGATCATGATAATGTACTTTCTTTCTGGGCTCCCTACAACAT
CCR5/2 GTGAGGCTCATTTTCGTGATCATGATAGTGACTTTCTTCTTCTGGGCTCCCTACAACAT

CCR5 CGTCCTTCTCCTGAGCACCTTCCAGAAATCCTTTGGCCTGGATCAATGCAGTAGCTCTA
CCR5/2 CGTCCTTCTCCTGAGCACCTTCCAGAAATCCTTTGGCCTGGATGACTTTGAGACCTCCA

CCR5 ATAGGCTGGACCAAGCCATGCAGGTGACAGAGACGCTTGGGATGACGCACTGCTGCATC
CCR5/2 GTCGGCTGGACCAAG**M**CATGCAGGTGACGGAGAC**W**CTTGGGATGACGCACTGCTGCATC

CCR5 AACCCCGTCATCTACGCCTTCGTGCGGGGAGAAGTTCAGAACTACCTCTTGGGGTATTT
CCR5/2 AACCCCATCATCTACGCCTTCGT**Y**GGTGAGAAGTTCAGAAGGTATCTCTCCATGTTCTT

CCR5 CCGAAAGCACATCGTCAGACGCTTTTGCAAAA---GCTGTCCCGTCTTCCAGGGAGAGG
CCR5/2 CCGAAAGCAC**R**TTGCCAAACATCTCTGCAACCTCGCTGCCAGTTTTCTGTGGGAAGA

CCR5 CTCCCAAGCGAGTAACTCCATTTATACCGGATCTACCGGCGAGCAGGAAATCTCTGTT
CCR5/2 CTGTAGAACGAGTGAGTTCAAGAAACCCCTTCCGCTGGGGAGCAGGAATCTCAATA

CCR5 GGCTTGTTGA
CCR5/2 GCT---TGA

Figure 3. Variable nucleotides (in bold) detected in the equine *CCR5* or *CCR5/2* genes. The predicted junction region between the *CCR5*-like and *CCR2*-like portions in the *CCR5/2* gene is underlined.

CCR5/2 mutations in the same 16 horse DNA samples. Thirteen SNPs were found. Seven of these SNPs changed the amino acid residue and one created an insertion/deletion (C/–) that generated both a frameshift and prematurely terminated translation. Alignment of the equine *CCR5* and *CCR5/2* sequences demonstrated that all of the

CCR5 SNP positions differed from those of the *CCR5/2* gene SNPs (Figure 3). This observation indicated that the individual SNPs could be genotyped within the horse *CCR5* gene using simple high-throughput assays. Each of the *CCR5* substitutions detected was subsequently genotyped by either RFLP or tetraprimer ARMS-PCR (Table 2).

Table 3. Major allele frequencies of synonymous SNPs within the coding region of the equine *CCR5* gene

Breed	N	63A	108C	267C	399C
American Saddlebred	26	0.79	0.85	0.79	0.98
Arabian	23	1.00	1.00	1.00	0.83
Quarter horse	10	0.95	0.95	0.95	0.85
Standardbred	22	0.95	0.95	0.95	0.95
Thoroughbred	73	0.81	0.82	0.79	0.95

N, number of horses studied.

The frequencies of identified alleles were estimated using DNA samples from 5 horse breeds (Table 3). Arabian horses were found to be polymorphic only for the C399T SNP, whereas both American Saddlebreds and Thoroughbreds demonstrated a high rate of variability in 3 of the 4 *CCR5* SNPs analyzed. Haplotype prediction using the 2SNP software (<http://alla.cs.gsu.edu/~software/2SNP/>) revealed 3 common and a single rare (<1%) SNP haplotypes spanning most of the horse *CCR5* gene (data not shown). Association analysis of haplotype markers rather than individual SNPs will facilitate future case/control studies investigating the genetic bases of horse susceptibility to infectious diseases.

Determination of Elephant and Chicken *CCR5* Gene Sequences

To obtain additional insights about *CCR5* and *CCR2* gene evolution, the sequences of these genes from 2 additional vertebrate species, elephant and chicken, were determined. Cattle, horse, and pig *CCR5* sequences were used to search the GenBank African savanna elephant (*Loxodonta africana*) genome trace archive database using the discontinuous Mega BLAST program. The genomic sequence 507006073

(G720P622497RG11.T0) was found to contain the 3' end of the elephant *CCR5* gene. The 5' end of this partial sequence was extended by genomic DNA walking to the ATG start codon. Subsequently, elephant genomic DNA and 2 *CCR5* specific primers (LaCCR5dnaF and LaCCR5dnaR; Table 1) were used to amplify and sequence the coding region of the elephant *CCR5* gene. The resulting sequence, which contained a 1062-bp ORF, was submitted to GenBank under accession number EF524204.

Comparison of the *CCR5* sequences from several species (Table 4) revealed a variation in ORF length ranging from 1056 bp (platypus; *Ornithorhynchus anatinus*) to 1071 bp (tenrec; *Echinops telfairi*). The length of a chicken (*Gallus gallus*) *CCR5* ORF in the GenBank database (NM_001045834) was substantially shorter (867 bp) than those of the orthologous mammalian genes. The NM_001045834 sequence was predicted from a genomic sequence of the chicken chemokine receptor cluster. To explore the possibility that this prediction was not accurate and that the length of the chicken *CCR5* ORF exceeded 867 bp, a search of the chicken GenBank EST database was performed using NM_001045834. Two ESTs, that did not overlap each other, were initially identified. The 3' part of the NM_001045834 sequence was found to be identical to the 5' part of EST AW239706 (reverse complement). This EST has a stop codon that is in frame with the NM_001045834 ORF as well as a 3' NCR of 292 bp. Another EST, BI393893, contained the 5' portion of the chicken *CCR5* mRNA, which consisted of a 257-bp 5' NCR and an additional in frame translation initiation codon located 201 bp upstream of the translation initiation codon predicted in the NM_001045834 sequence. When the chicken GenBank database was searched with the

Table 4. Lengths of ORFs and GenBank accession numbers of selected vertebrate *CCR5*, *CCR2*, and *CCR3* genes

Vertebrate species	CCR5		CCR2		CCR3	
	ORF ^a	GenBank	ORF ^a	GenBank	ORF ^a	GenBank
Horse	1059	DQ629590	1143	EF649735	1080	XM_001496014
Cat	1059	NM_001009248	1122	NM_001097580	1086	AANG01259742
Dog	1059	NM_001012342	1119	XM_541906	1080	NM_001005261
Cattle	1059	NM_001011672	1113	XM_584158	1077	XM_869148
Domestic pig	1059	NM_001001618	1122	NM_001001619	1077	NM_001001620
Elephant	1062	EF524204	N/A	N/A	1083	AAGU01193615
Human	1059	NM_000579	1083	XM_001133431	1068	NM_001837
Chimpanzee	1059	AF005663	1083	XM_001149592	1068	XM_526193
Rhesus	1059	MMU77672	1083	NM_001032806	1068	NM_001047140
Rabbit	1059	DQ444458	1080	AAGW01631601	1080	EU250018
Mouse	1065	MMU68565	1122	NM_009915	1080	NM_009914
Rat	1065	NM_053960	1122	NM_021866	1080	NM_053958
Guinea pig	1062	AAKN01327787	1056	AAKN01088403	1077	AF060698
Armadillo	N/A	N/A	1104	AAGV01675368	1011	AAGV01340262
Tenrec	1071	AAIY01265902	N/A	N/A	1086	AAIY01212007
Opossum	1059	XM_001379584	1083	XM_001369297	1065	XM_001379572
Platypus	1056	AAPN01046046	1047	XM_001517237	1137	XM_001513744
Chicken	867 ^b	NM_001045834	1065	NM_001045835	N/A	N/A
Lizard	1068	AAWZ01026126	1074	AAWZ01026126	N/A	N/A

^a Length of ORF in base pairs.

^b Found to be 1068 bp in the present study.

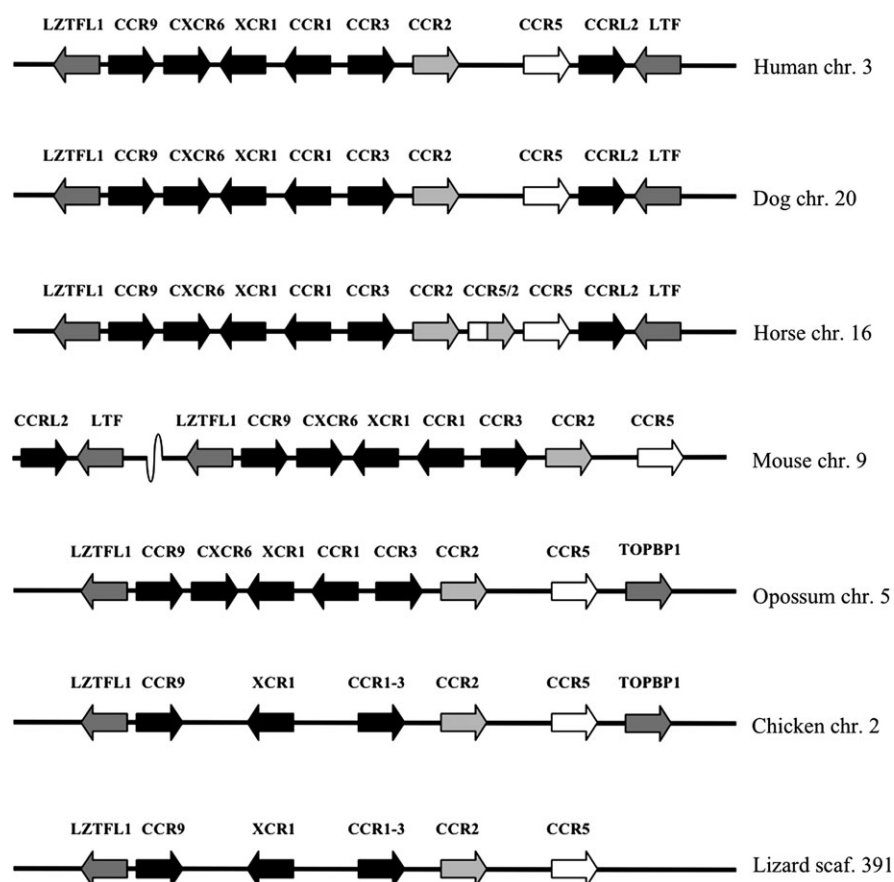


Figure 4. Maps of vertebrate *CCR* gene clusters compiled from the whole-genome sequencing projects represented at the UCSC web site (<http://genome.ucsc.edu>). Arrows indicate direction of RNA transcription. The *CCR2* genes are indicated by light gray, the *CCR5* genes by white, and the other *CCR* genes by black arrows. The genes flanking the *CCR* cluster are indicated in dark gray. The diagrams shown are not to scale.

BI393893 sequence, 2 additional short ESTs, CK613374 and DT658610, were detected. Neither CK613374 nor DT658610 overlapped the NM_001045834 sequence but had translation initiation codons in positions that were similar to that in the BI393893 sequence. Of the three 5' terminal chicken *CCR5* ESTs detected in GenBank, the DT658610 sequence had the longest 5' NCR (314 bp). Two primers, GgCCR5rnaF and GgCCR5rnaR (Table 1), designed based on the DT658610 or AW239706 sequences, respectively, were used to amplify and sequence the *CCR5* transcript from a chicken cDNA (intestine). The resulting full-length mRNA sequence was submitted to GenBank under accession number EF524205. This sequence consisted of a 5' NCR of 247 bp, an ORF of 1068 bp, and a 3' NCR of 118 bp. It is expected that neither the 5' nor 3' NCR sequences in the EF524205 sequence is complete.

Syntenic and Evolution of the Vertebrate *CCR2* and *CCR5* Genes

The *CCR2* and *CCR5* genes are members of a cluster of 8 chemokine receptor genes that are highly conserved among

mammals. In the human genome, this cluster is located in a 500-kb region of chromosome 3 between the *LZTFL1* and *LTF* genes (Figure 4). A search of the UCSC genome browser revealed that the gene order in the corresponding region of dog chromosome 20 is identical to that in humans. In addition to the *CCR* genes that were identified in the human and dog clusters, a unique equid *CCR5/2* gene was found in the horse (chromosome 16), donkey, and zebra genomes (see above). Mouse and rat have identical *CCR* gene orders but the rodent *CCRL2* gene has been translocated to a different region of the same chromosome (chromosome 8 and 9 in rat and mouse, respectively). In opossum (*Monodelphis domestica*) and chicken, the *CCR* cluster is located in chromosome 5 and 2, respectively, but the *CCRL2* gene is absent. Also, there is a single ortholog of the *CCR1* and *CCR3* genes in both the chicken and lizard (*Anolis carolinensis*) genomes (Figure 4). In the frog (*Xenopus tropicalis*), only one ortholog of the mammalian *CCR5* and *CCR2* genes was found within genomic scaffold_36 (data not shown), suggesting that tandem duplication of these 2 genes occurred after divergence of amphibians but before the split of the reptilian, avian, and

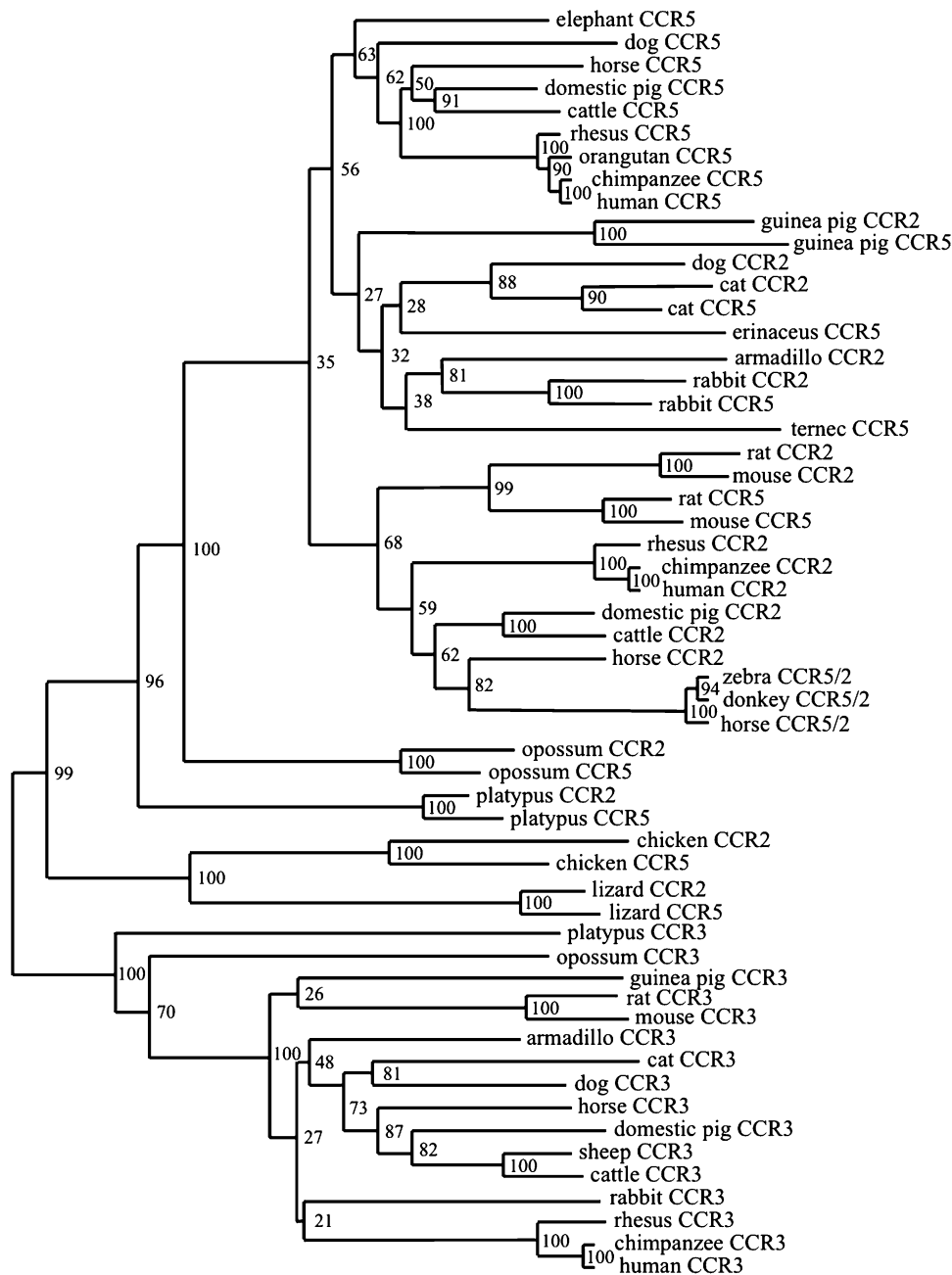


Figure 5. Phylogenetic tree of selected vertebrate *CCR2*, *CCR5*, and equid *CCR5/2* genes. The sequences of mammalian *CCR3* genes were used as the outgroups. The numbers at the tree nodes indicate the corresponding bootstrap values estimated by 1000 bootstrap replications.

mammalian lineages from a common ancestor (DeVries et al. 2006).

To analyze the relationships of the *CCR2* and *CCR5* genes, a phylogenetic tree was constructed using the closest paralog, the *CCR3* gene for each species, as an outgroup. For several mammalian species, the orthologous *CCR5* genes grouped in a separate subtree (Figure 5). However, the lizard, chicken, opossum, platypus, rabbit, guinea pig (*Cavia porcellus*), cat, and rodent *CCR2* and *CCR5* genes, but not

the paralogous mammalian *CCR3* genes, were present in species-specific clusters in the tree. This clustering contradicted the paralogous nature of these genes as inferred from their chromosomal location and suggested intensive concerted evolution of these genes by gene conversion. Evidence of concerted evolution of the *CCR2* and *CCR5* genes was previously reported in mice (Shields 2000), leporids (Carmo et al. 2006) and felids (Esteves et al. 2007; Vazquez-Salat et al. 2007).

Table 5. Divergent and convergent substitution rates in paralogous and orthologous lineages of *CCR2/CCR5* and *CCR2/CCR3* gene pairs

Pairs of species	CCR2 and CCR3 genes			CCR2 and CCR5 genes		
	Divergent substitutions/gaps	Convergent substitutions		Divergent substitutions/gaps	Convergent substitutions	
		Paralogous	Orthologous		Paralogous	Orthologous
Mouse/rat	323/92	5	7	117/88	9	0
Human/mouse	167/41	16	10	49/44	47	10
Human/horse	187/40	22	23	73/48	23	5
Human/opossum	103/22	22	23	8/18	106	2

To compare the evolution rates in both paralogous and orthologous genes, the distribution of divergent and convergent substitutions among various lineages was analyzed as previously described (Perelygin et al. 2006b). The goal of this analysis was to estimate the background level of convergent substitution between evolutionarily unrelated lineages (different genes in different species) and to compare it to the corresponding occurrence of convergent substitutions in paralogous genes (different genes in the same species). If the paralogous genes evolved independently from each other, their convergent substitution rates would be equal to those within orthologous genes. The results of this analysis (Table 5) demonstrated that the rate of convergent substitution was approximately the same between paralogous and orthologous *CCR2* and *CCR3* genes indicating the absence of conversion between these 2 genes. However, the convergent substitution rate between the paralogous *CCR2* and *CCR5* genes was significantly higher than the corresponding rate between the orthologous genes. For the human–mouse and human–opossum comparisons, the number of convergent substitutions was close to or exceeded the number of divergent substitutions suggesting that the duplication and divergence of *CCR2* and *CCR5* genes occurred prior to mammalian speciation. When gene divergence occurs prior to speciation, the phylogenetic analysis cannot correctly infer the true relationship between genes and species. This was the major reason that species-specific clustering of the rodent, opossum, and many other paralogous genes was observed in the tree.

The hypothesis that *CCR2* and *CCR5* gene duplication occurred prior to speciation is supported by the increased similarity of the 5' ends of the orthologous gene sequences in the alignment that was used to build the phylogenetic tree. The majority of the divergent gaps are located at the beginning of paralogous genes (data not shown). The alternative hypothesis that species divergence occurred prior to gene duplication would assume an independent origin of each of the gaps in several lineages. Because gap generation is a relatively rare evolutionary event and gene duplication would have had to occur independently many times in the lizard, chicken, platypus, opossum, rodent, and primate lineages, this alternative hypothesis seems improbable.

The region-specific patterns of gene conversions were reported in several other gene families (Higgs et al. 1984; Zhao et al. 1998; Lazzaro and Clark 2001; Archibald and Roger 2002a, 2002b; Bettencourt and Feder 2002; Desjardins

et al. 2002; Drouin 2002; Israel et al. 2002; Nagawa et al. 2002; Annilo et al. 2003; Perelygin et al. 2006b; Kruithof et al. 2007). To study the distribution of gene conversion events along the *CCR5* and *CCR2* genes, we applied the Kolmogorov–Smirnov test as described previously (Perelygin et al. 2006b). In this approach, the distribution of nucleotide substitutions between a pair of paralogous sequences is compared with that in a pair of orthologous sequences. A gene conversion event results in a high similarity between affected parts of “donor” and “acceptor” genes and a relatively high divergence between the remaining parts of the 2 genes. Therefore, in the case of a recent gene conversion in a particular species, the distributions between paralogous genes will deviate from the distributions between orthologous genes, which can be detected statistically by the nonparametric Kolmogorov–Smirnov test. The most significant deviation was observed for each of the species that form clusters of paralogous *CCR2* and *CCR5* genes in the tree (Figure 5): lizard ($P = 10^{-47}$), chicken ($P = 10^{-38}$), guinea pig ($P = 10^{-28}$), platypus ($P = 10^{-12}$), cat and rodents ($P = 10^{-9}$), opossum, rabbit, and rhesus monkey ($P = 10^{-6}$). In all these cases, the observed deviations can be explained by one or several recent gene conversion events affecting large portions of the genes. Actually, all regions of these genes with the exception of their 5' ends were involved in gene conversion in different species. The 5' ends differ significantly between *CCR2* and *CCR5* genes and can serve as reliable markers in combination with gene order for establishing orthologous relationships between these genes in various species.

For most of the other placental mammals, deviations in the distributions of nucleotide substitutions were not as significant ($P < 10^{-6}$) as the deviations reported above. Therefore, the correct topology of the phylogenetic tree would not be substantially affected for these mammals. However, comparison of the numbers of convergent substitutions between paralogous and orthologous genes (Table 5) revealed significant gene conversion in these species suggesting that gene conversion is an ongoing process that randomly affects small portions of the *CCR2* and *CCR5* genes in a number of mammalian species.

Origin and Evolution of the Equid *CCR5/2* Gene

Among the species for which genomes have been sequenced to date, the equids are the only ones with an additional

CCR5/2 gene. The *CCR5/2* gene likely originated from an unequal crossing over event between the *CCR2* and *CCR5* genes (Figure 2A) and would be expected to have a *CCR5*-like 5' end and a *CCR2*-like 3' end. To confirm this assumption and to identify the most probable recombination site, the cumulative distributions of nucleotide differences were calculated between 3 pairs of genes: *CCR2* and *CCR5* (S_{25}), *CCR2* and *CCR5/2* (S_{2x}), and *CCR5* and *CCR5/2* (S_{5x}). The recombination site was located using break-points in the lines of a phase diagram of the distributions for S_{2x} or S_{5x} plotted versus the distribution for S_{25} (Figure 2B). The break point was estimated to be located between positions 394 to 402 bp in the *CCR5/2* gene. This region encodes the inner loop located between the TM3 and TM4 domains. The *CCR5*-like portion of the chimeric protein contains the N-terminal sequence and the first 3 TM domains. The *CCR2*-like portion includes the remaining 4 TM domains and the C-terminal domain. Because the *CCR2*-like portion is larger, the equid *CCR5/2* gene clustered with the horse *CCR2* gene in the tree (Figure 5). The presence of the divergent *CCR5*-like portion increased the length of the *CCR5/2* branch of the tree.

As an additional means of characterizing *CCR5/2* gene evolution, the number of synonymous and nonsynonymous substitutions was counted separately in the *CCR5* and *CCR2* portions of the chimeric gene (Figure 6). Although the number of synonymous substitutions in *CCR5/2* gene was close to that in the corresponding parts of the *CCR5* and *CCR2* genes, the number of nonsynonymous substitutions in the chimeric gene was about 3–4 times higher than those in the original genes. The *CCR5/2* gene may have evolved without any functional restrictions or under selection that supports nonsynonymous substitutions in functional domains. The distribution of the nonsynonymous substitutions along the sequence of the *CCR5/2* gene was uniform and did not appear to be associated with any known functional sites, suggesting that there are no functional restrictions on the evolution of the *CCR5/2* gene. It is likely that the horse *CCR5/2* protein currently has no specific functional activity and that both the *CCR2* and *CCR5* proteins perform their regular cellular functions.

Similar numbers of accumulated substitutions were found in the human and horse lineages: 75.4 ± 8.7 and 71.0 ± 8.4 , respectively, in the *CCR5* genes and 84.3 ± 9.2 and 89.7 ± 9.5 , respectively, in *CCR2* genes. These calculations suggested similar substitution rates in ungulates and primates after their divergence from a common ancestor and made it possible to estimate the time when recombination between the horse *CCR2* and *CCR5* genes had occurred. According to the data shown in Figure 6, the total number of synonymous and nonsynonymous substitutions between the horse *CCR2* and *CCR5* genes prior to origination of the *CCR5/2* gene was 46 (8 + 10 + 12 + 16). After the unequal crossing over event between the *CCR2* and *CCR5* genes, 32 (7 + 6 + 11 + 8) substitutions accumulated. The divergence between primates and ungulates was estimated to occur about 95 million years ago (Ma) (Soligo et al. 2007). Therefore, the age of the horse

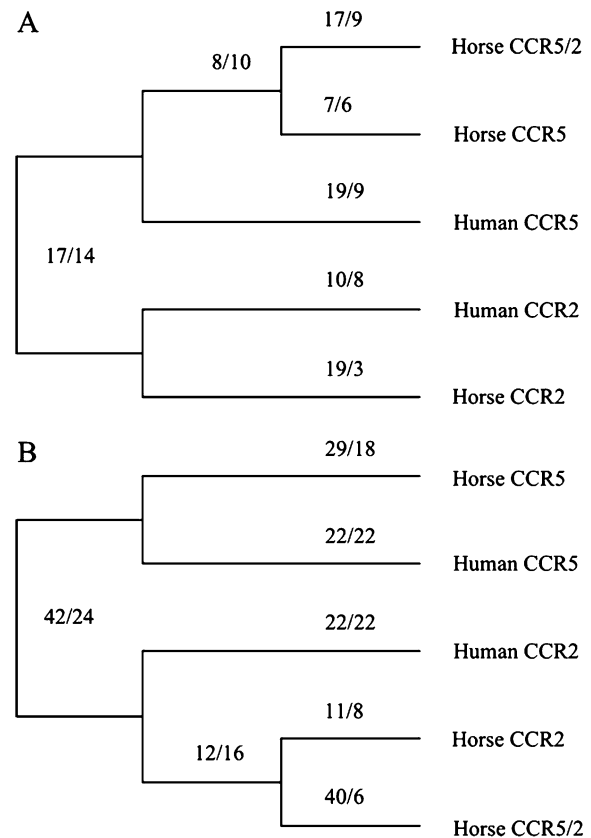


Figure 6. Ratios of nonsynonymous to synonymous substitutions in the horse *CCR5/2* gene compared with the human and horse *CCR5* and *CCR2* genes. (A) Tree corresponding to the *CCR5*-like portion (positions from 1 to 396 bp). (B) Tree corresponding to the *CCR2*-like portion (positions from 397 to 1068 bp) of the *CCR5/2* gene.

CCR5/2 gene is estimated to be about $95 \times 32/(32 + 46) = 39$ Ma.

Conclusions

Evidence for concerted evolution described in this study provides an explanation for the high sequence similarity between the vertebrate *CCR2* and *CCR5* genes. Special precautions are therefore required to discriminate the sequences of these paralogous genes in DNA samples, and it also likely that antibodies made to either *CCR2* or *CCR5* proteins would cross-react. This would also be the case for the *CCR5/2* gene and gene product in equids. Although not yet experimentally confirmed, the biological properties (e.g., antiviral activities) of individual CCR proteins previously observed in particular vertebrate lineages (e.g., humans and mice) may also be conserved in other species (e.g., equids). The resequencing of the *CCR5* genes in a limited number of control horses revealed synonymous but no nonsynonymous or missense mutations. Sequence

analysis of much larger numbers of DNA samples, especially those from WNV-infected horses that develop CNS disease, may reveal a low-frequency mutation similar to the rare human *CCR5Δ32* mutation that inactivates the equine CCR5 protein function and increases susceptibility to infectious disease. The SNPs identified will be useful for developing a horse haplotype map.

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